

6CCS3PRJ

Designing for Consent Revocation in Online Platforms: WhatsApp Photo Sharing

Final Project Report — Appendices

Author: Mert Ayranci

Supervisor: Dr. Kovila Coopamootoo

Student ID: 21175534

April 20, 2026

About this document. This file contains the appendices that accompany the main project report. It is submitted as a separate PDF in line with the university's submission instructions. Cross-references of the form "Section X" or "Chapter X" refer to the main report unless stated otherwise.

Contents of appendices

A Thematic Analysis Materials **3**
A.1 Interview extracts board (raw) . . . 4
A.2 Inductive codebook (initial iteration) . . . 5
A.3 Interview extracts (refined subset) . . . 7
A.4 Working codebook (simplified) . . . 8
A.5 Quote-to-code coding sheet . . . 9
A.6 Final themes table . . . 10

B Interview Guide **11**
B.1 Section 0: Mandatory consent statement . . . 12
B.2 Section 1: Warm-up and current photo-sharing behaviour . . . 12
B.3 Section 2: (Scenario 1) Before sending a photo . . . 13
B.4 Section 3: (Scenario 2) Changing permissions after sending . . . 14
B.5 Section 4: (Scenario 3) Recipient view, request access, and fairness . . . 14
B.6 Section 5: (Scenario 4) Forwarding and residual concerns . . . 15
B.7 Section 6: (Scenario 5) Co-subject request for revocation . . . 16
B.8 Section 7: Wrap-up and final reflections . . . 16
B.9 Interview scenario slides (stimulus material) . . . 17

C User Guide **20**
C.1 Purpose . . . 20
C.2 Prerequisites . . . 20
C.3 Install and run . . . 20
C.4 Optional: production build . . . 21
C.5 Feature checklist (where to see each behaviour) . . . 21
C.6 Optional: automated tests . . . 22

D Ethical clearance **23**

E Source Code **32**
E.1 Submission contents . . . 32
E.2 Repository layout . . . 32
E.3 File index . . . 33
E.4 Notes . . . 37

Appendix A

Thematic Analysis Materials

This appendix reproduces the reflexive thematic analysis workbook as exported from the project Miro board.

A.1 Interview extracts board (raw)

First page of the full colour-coded transcript extract board.



A.2 Inductive codebook (initial iteration)

First page of the initial 56-code codebook produced through close reading of the interview transcripts.


Control and Agency
Understandibility and Clarity
Trust and Transparency
Social Emotional Impact
Forwarding/Redistribution
Co-Subject Multi-Party Privacy
UI Design Feedback

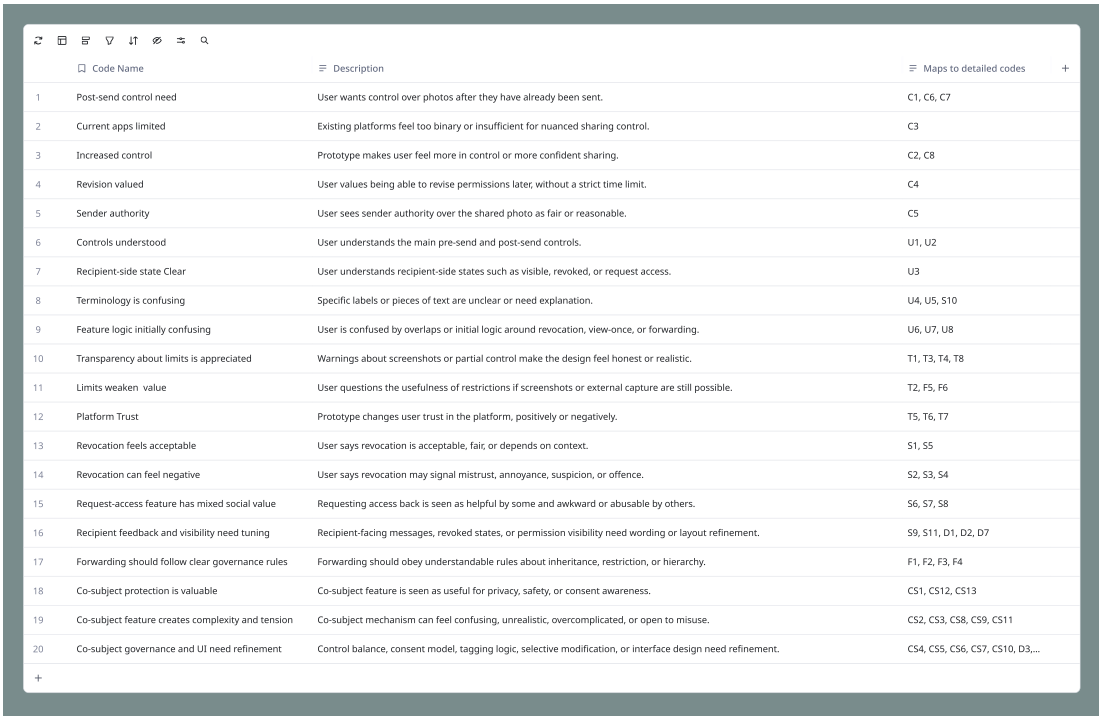
A.3 Interview extracts (refined subset)

First page of the refined extract set used for focused coding.



A.4 Working codebook (simplified)

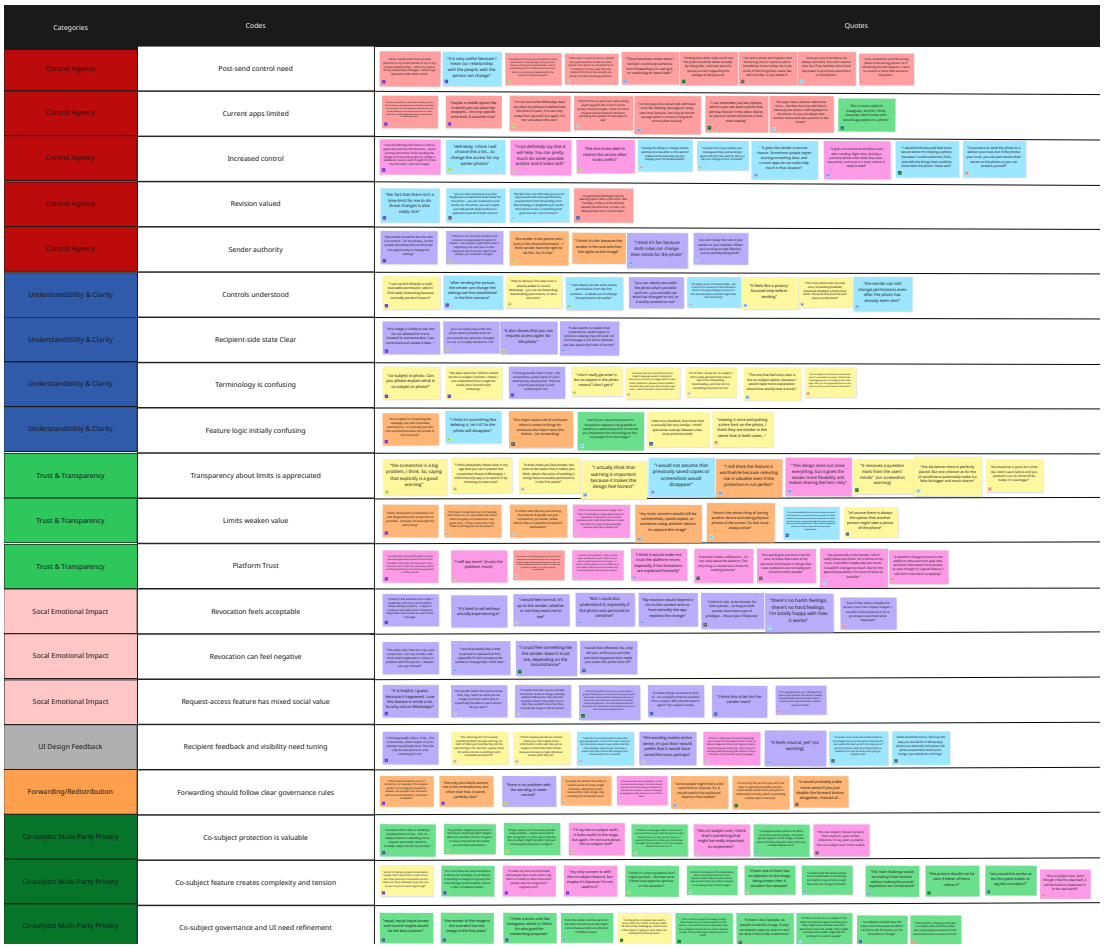
First page of the simplified 20-code working codebook used for cross-case tagging.



	Code Name	Description	Maps to detailed codes
1	Post-send control need	User wants control over photos after they have already been sent.	C1, C6, C7
2	Current apps limited	Existing platforms feel too binary or insufficient for nuanced sharing control.	C3
3	Increased control	Prototype makes user feel more in control or more confident sharing.	C2, C8
4	Revision valued	User values being able to revise permissions later, without a strict time limit.	C4
5	Sender authority	User sees sender authority over the shared photo as fair or reasonable.	C5
6	Controls understood	User understands the main pre-send and post-send controls.	U1, U2
7	Recipient-side state Clear	User understands recipient-side states such as visible, revoked, or request access.	U3
8	Terminology is confusing	Specific labels or pieces of text are unclear or need explanation.	U4, U5, S10
9	Feature logic initially confusing	User is confused by overlaps or initial logic around revocation, view-once, or forwarding.	U6, U7, U8
10	Transparency about limits is appreciated	Warnings about screenshots or partial control make the design feel honest or realistic.	T1, T3, T4, T8
11	Limits weaken value	User questions the usefulness of restrictions if screenshots or external capture are still possible.	T2, F5, F6
12	Platform Trust	Prototype changes user trust in the platform, positively or negatively.	T5, T6, T7
13	Revocation feels acceptable	User says revocation is acceptable, fair, or depends on context.	S1, S5
14	Revocation can feel negative	User says revocation may signal mistrust, annoyance, suspicion, or offence.	S2, S3, S4
15	Request-access feature has mixed social value	Requesting access back is seen as helpful by some and awkward or abusable by others.	S6, S7, S8
16	Recipient feedback and visibility need tuning	Recipient-facing messages, revoked states, or permission visibility need wording or layout refinement.	S9, S11, D1, D2, D7
17	Forwarding should follow clear governance rules	Forwarding should obey understandable rules about inheritance, restriction, or hierarchy.	F1, F2, F3, F4
18	Co-subject protection is valuable	Co-subject feature is seen as useful for privacy, safety, or consent awareness.	CS1, CS12, CS13
19	Co-subject feature creates complexity and tension	Co-subject mechanism can feel confusing, unrealistic, overcomplicated, or open to misuse.	CS2, CS3, CS8, CS9, CS11
20	Co-subject governance and UI need refinement	Control balance, consent model, tagging logic, selective modification, or interface design need refinement.	CS4, CS5, CS6, CS7, CS10, D3,...

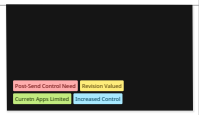
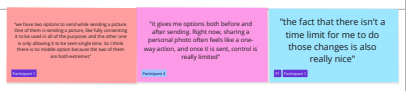
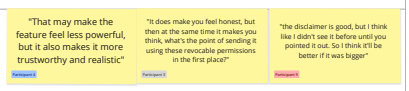
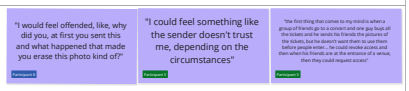
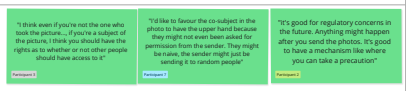
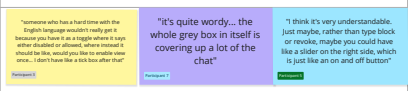
A.5 Quote-to-code coding sheet

First page of the quote-to-code mapping sheet.



A.6 Final themes table

First page of the consolidated themes table (definitions, codes, and example quotations).

Theme	Description	Codes	Example Interview Quotes
Theme 1: From Binary to Revocable Consent	This theme concerns participants' views on current photo-sharing controls as too limited or too extreme. It captures their preference for permission settings that can be adjusted both before and after sending, rather than fixed at the moment of disclosure.		
Theme 2: Bounded Control and Honest Limitations	This theme concerns how participants responded to the system's communication of its own limits, particularly around screenshots and saved copies. It captures the tension between valuing transparency and questioning the usefulness of controls that cannot offer complete protection.		
Theme 3: Revocation as Social Negotiation	This theme concerns how participants interpreted photo revocation as an interpersonal action rather than a purely technical one. It includes responses shaped by trust, fairness, awkwardness, and views on the request-access mechanism.		
Theme 4: Co-Subject Privacy as a Multi-Consent Problem	This theme concerns participants' reactions to giving depicted individuals a role in post-send control. It includes both support for the underlying privacy principle and concerns about feasibility, power balance, and how such control should be governed.		
Theme 5: Interface Legibility and Permission Understandibility	This theme concerns how clearly the interface communicated permission states, feature logic, and available actions. It includes participant responses to wording, repetition, layout, and the balance between visibility and message-space efficiency.		

Appendix B

Interview Guide

This appendix records the scenario-based moderator script used in the evaluation interviews. The wording follows the version spoken in sessions (including the consent statement and scenario prompts). Each numbered question lists the **linked research questions** (Main RQ and RQ1–RQ5) taken from the analysis mapping used alongside the guide: RQ1 addresses current platform limitations; RQ2 expectations and post-sharing concerns; RQ3 clarity of the interface and permission model; RQ4 fairness and social implications; RQ5 trust, safety, control, and realism of revocation. The main research question is stated in Chapter 1.

Research question key

Main RQ

Overall contribution of the proposed design.

RQ1

Limitations of current platforms for controlling shared photos.

RQ2

User expectations, practices, and concerns about post-sharing control.

RQ3

Clarity and understandability of the interface and permission model.

RQ4

Social implications for stakeholders.

RQ5

Trust, safety, perceived control, and realism of revocation.

B.1 Section 0: Mandatory consent statement

Say verbatim:

> "Before we begin, I'd like to confirm your consent to participate in this interview. This interview is part of my research on how messaging platforms might support consent revocation for shared photos. The purpose of this interview is to understand your views and reactions to a design concept. During the interview, I will show you example screens from a prototype. I am not testing you — I am testing the design. There are no right or wrong answers, and honest reactions are very helpful. Some questions are about privacy, control, and what happens to photos after they are shared. You can skip any question or stop the interview at any time. With your permission, I would like to audio-record this interview so I can analyse the responses later. The recording will only be used for this project, and your responses will be anonymised. You can choose to stop the recording or withdraw from the interview at any time. If you're happy to proceed, could you please state your full name and age, and confirm that you agree to be interviewed and recorded?"

B.2 Section 1: Warm-up and current photo-sharing behaviour

1. **Can you tell me a bit about how you usually share photos on apps like WhatsApp?**

   Linked: RQ2.

   Probes optional: private vs group chats; casual vs personal photos; frequency.

2. **When you send a photo to someone, do you usually think about what might happen to it afterwards? If yes, what kinds of concerns or challenges come to mind?**

   Linked: RQ2.

   Probes optional: saving, forwarding, screenshots, long-term retention; active vs incidental

thought.

3. **Have you ever wanted more control over a photo after you had already sent it?**

   Linked: RQ1; RQ2.

   Probes optional: type of control wished for; whether the app helped.

4. **What do you think current apps do well or badly when it comes to controlling shared photos?**

   Linked: RQ1.

B.3 Section 2: (Scenario 1) Before sending a photo

Show: photo composer / permission setup (view-once, forwarding, download, co-subject, time limit; note on screenshots and saves).

5. **What do you think is happening on this screen?**

   Linked: RQ3.

   Probes optional: purpose of controls; what the sender decides.

6. **What do you think each of these settings means?**

   Linked: RQ3.

   Probes optional: view once; forwarding disabled; download disabled; time limit; co-subject in photo.

7. **Which of these options feel useful to you, and which feel unnecessary or unclear?**

   Linked: RQ2; RQ3.

   Probes optional: realistic pre-send use; effort; missing options.

8. **There is also a warning here saying that screenshots and saves cannot be undone. How do you feel about that?**

   Linked: RQ3; RQ5.

   Probes optional: honesty vs usefulness; whether the warning is wanted.

Moderator note: Only explain controls if the participant is completely stuck; prior interpretation is the primary data.

B.4 Section 3: (Scenario 2) Changing permissions after sending

Show: sent photo with photo menu (revoke/restore, forwarding, download, time limit).

9. **Looking at this screen, what do you think the sender is able to do now?**

   Linked: RQ3.

   Probes optional: what is changeable after send; comparison to deleting a message; revoke vs permission change.

10. **What would you expect to happen if the sender revokes access here?**

    Linked: RQ3; RQ5.

    Probes optional: visibility; recipient notification; offline copies; if already seen.

11. **How understandable does this post-send control feel to you?**

    Linked: RQ3.

    Probes optional: clarity of change; menu affordances; discoverability.

12. **Would this kind of feature make you feel more in control after sending a photo? Why or why not?**

    Linked: RQ5.

    Probes optional: trust; anxiety; realistic use.

Moderator note: Keep focus on clarity, expectations, realism, and perceived control rather than implementation detail.

B.5 Section 4: (Scenario 3) Recipient view, request access, and fairness

Show: recipient-side revoked placeholder (screenshots / prior viewing note).

13. **What do you think this screen is telling the recipient?**

    Linked: RQ3; RQ4.

    Probes optional: clarity of change; sufficiency of explanation.

14. **How would you feel if you were the recipient in this situation?**

Linked: RQ4; RQ5.

Probes optional: reasonable vs awkward; fairness tone.

15. **What do you think about the idea that the recipient can request access again?**

    Linked: RQ4.

    Probes optional: comfort as sender or recipient; balance.

16. **Does this feel fair to both people, or more fair to one side than the other?**

    Linked: RQ4.

    Probes optional: sender power; explanation; objection rights.

17. **What do you think about the wording used here?**

    Linked: RQ4.

    Probes optional: neutrality vs blame.

Moderator note: Do not defend the revoked state; negative reactions are informative.

B.6 Section 5: (Scenario 4) Forwarding and residual concerns

Show: forwarding behaviour and secondary recipient where relevant; request-access flow and sender accept/reject if shown.

18. **What do you think this forwarding feature is trying to do?**

    Linked: RQ3.

    Probes optional: blocked vs allowed forwarding; sender rules after forward.

19. **How do you feel about the idea that the sender's rules continue even after the photo is forwarded?**

    Linked: RQ3; RQ5.

    Probes optional: usefulness; realism; over-control.

20. **Even with these features, what concerns would still remain for you?**

    Linked: RQ1; RQ2; RQ5.

    Probes optional: screenshots; prior downloads; residual value of imperfect control.

B.7 Section 6: (Scenario 5) Co-subject request for revocation

Show: co-subject notification and request flow; sender accept/reject.

21. **What do you think about that idea?**

    Linked: RQ4; RQ5.

    Probes optional: fairness; invasiveness; complexity.

22. **For this kind of scenario where the photo includes the sender and other co-subjects, who do you think should have the strongest control in that situation: the sender, the person in the photo, both equally, or someone else?**

    Linked: RQ4.

23. **How realistic does this feel in a real messaging app?**

    Linked: RQ5.

Moderator note: Use only if time allows or if co-subject control is a session focus.

B.8 Section 7: Wrap-up and final reflections

25. **Would this make you feel more in control when sharing personal photos? Why or why not?**

    Linked: RQ5.

26. **Would it make you trust the platform more, less, or no differently?**

    Linked: RQ5.

27. **What part felt most clear or useful?**

    Linked: RQ3; RQ5.

28. **What part felt most confusing, unrealistic, or uncomfortable?**

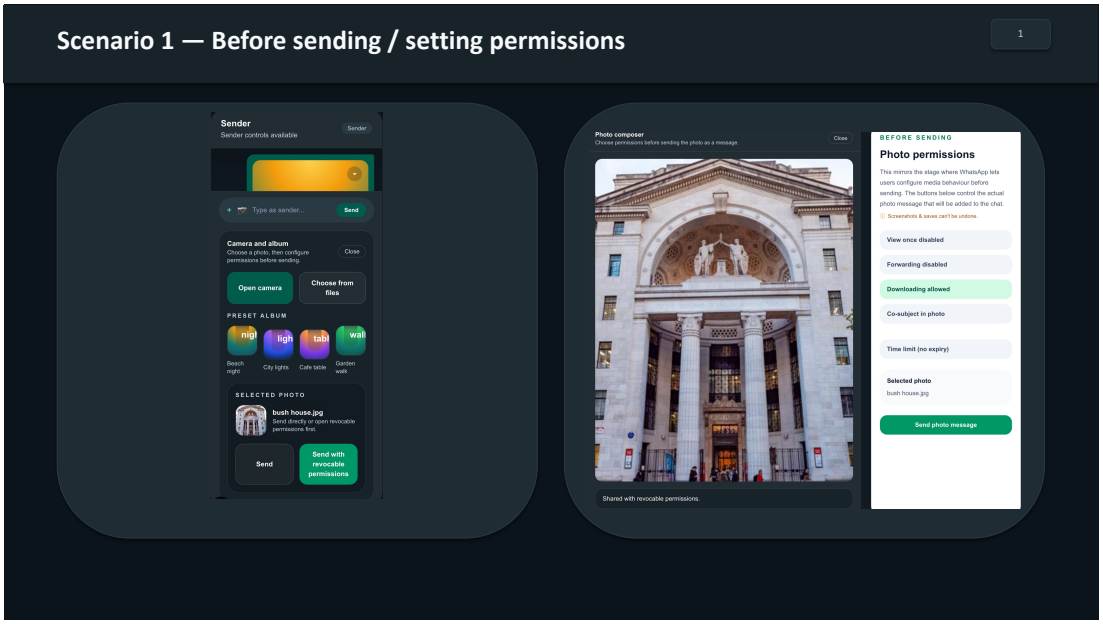
    Linked: RQ1; RQ3; RQ5.

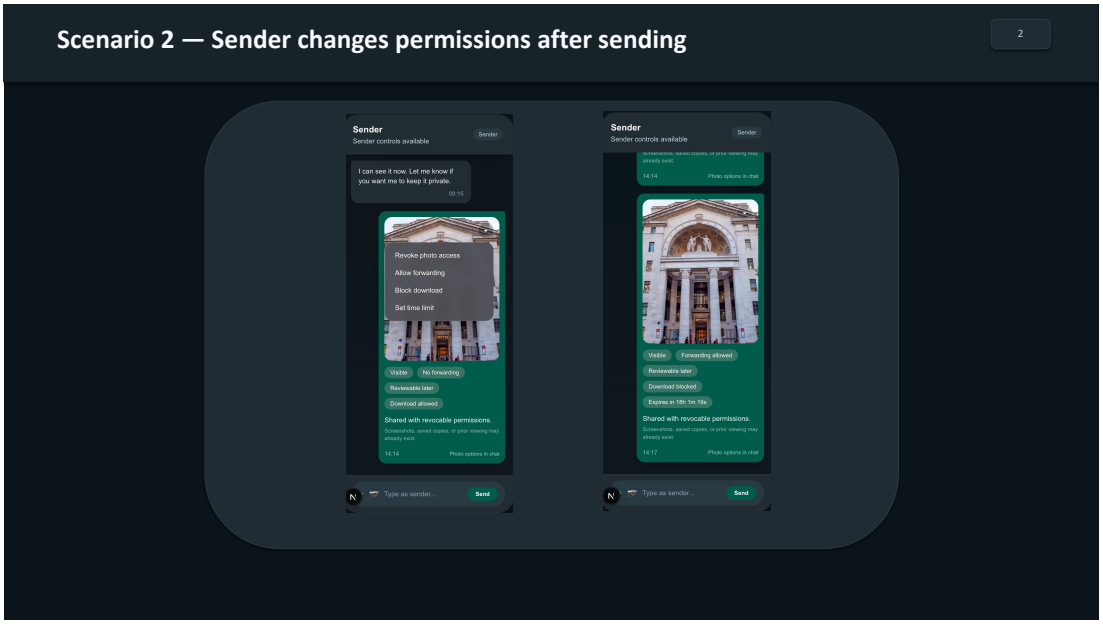
29. **From 1 to 5, how clear do you think everything feels?**

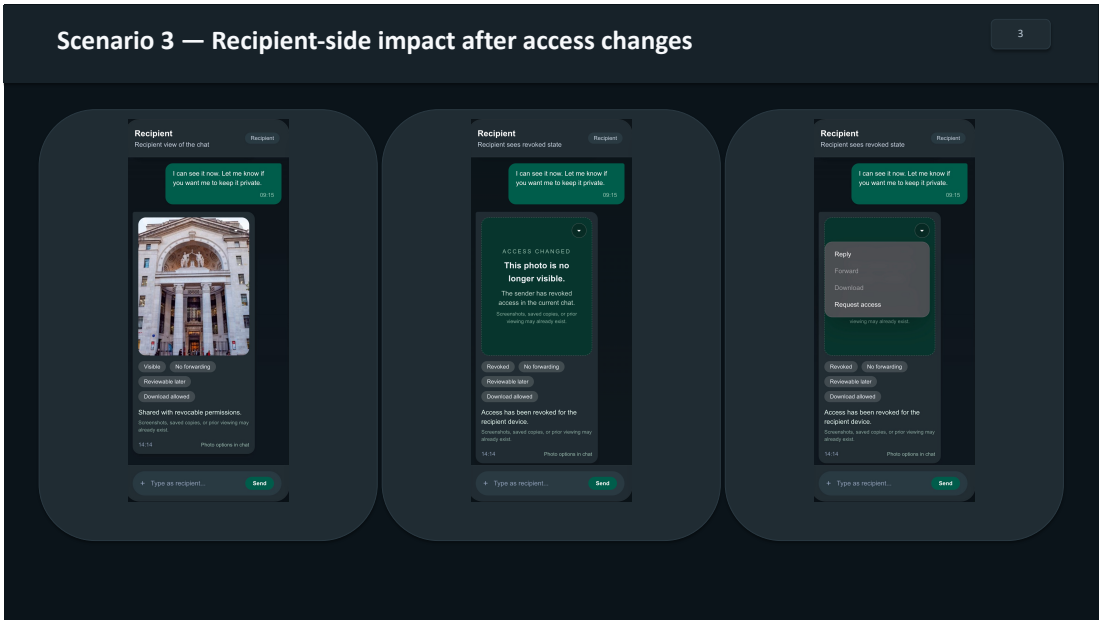
    Linked: RQ3.

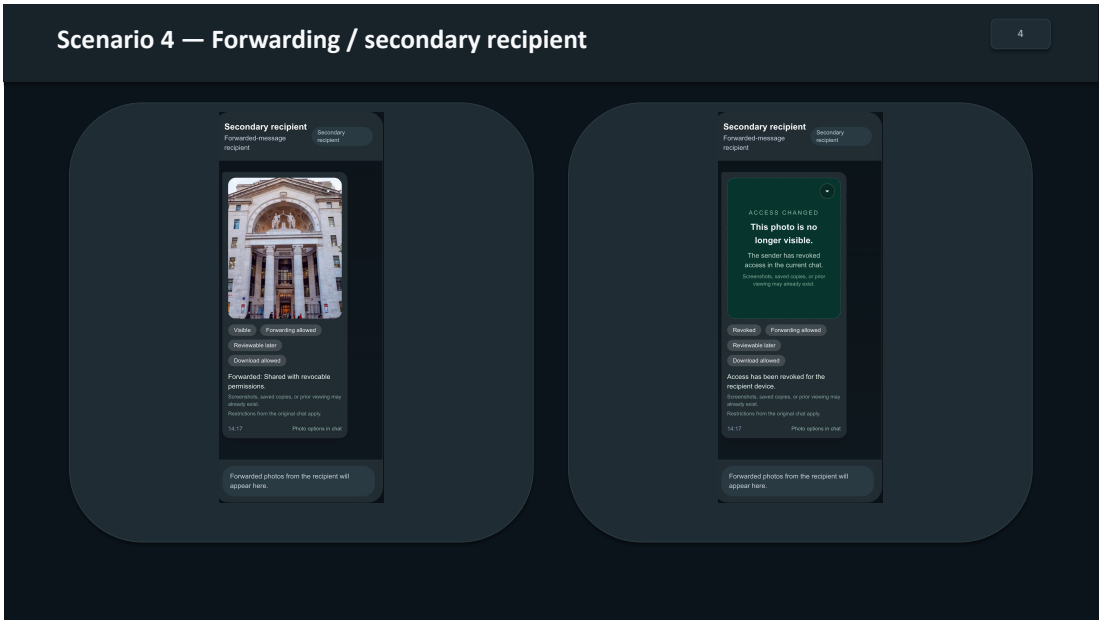
B.9 Interview scenario slides (stimulus material)

This section reproduces the slide deck *Consent Revocation Interview Scenarios.* The screenshots embedded in these slides were taken from the **first round** of the prototype implementation, before later visual or interaction refinements and they are preserved here to document the materials participants actually saw.



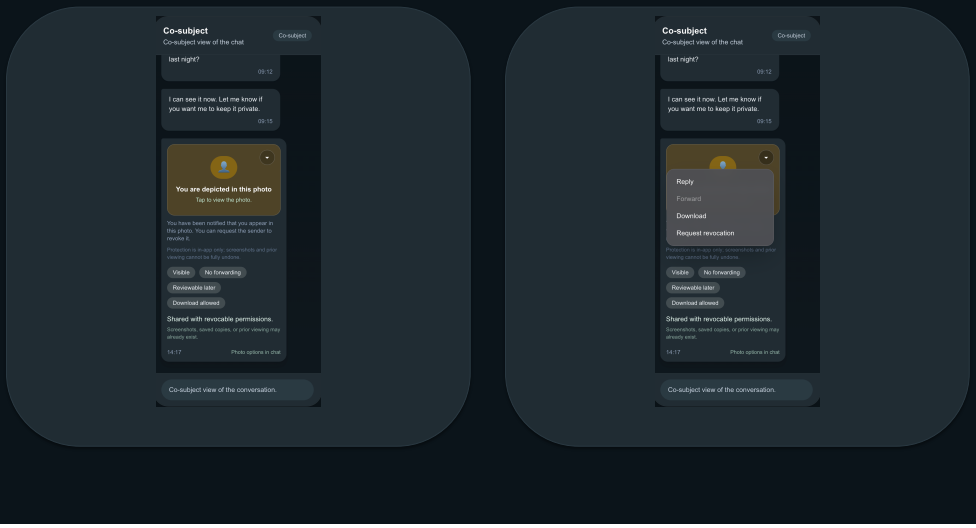



Scenario 3 — Recipient-side impact after access changes
3
Recipient
Recipient view of the chat
Recipient
I can see it now. Let me know if you want me to keep it private.
09:15
Visible
No forwarding
Reviewable later
Download allowed
Shared with revocable permissions.
Screenshots, saved copies, or prior viewing may already exist.
14:14
Photo options in chat
+ Type as recipient...
Send
Recipient
Recipient sees revoked state
Recipient
I can see it now. Let me know if you want me to keep it private.
09:15
ACCESS CHANGED
This photo is no longer visible.
The sender has revoked access in the current chat.
Screenshots, saved copies, or prior viewing may already exist.
Revoked
No forwarding
Reviewable later
Download allowed
Access has been revoked for the recipient device.
Screenshots, saved copies, or prior viewing may already exist.
14:14
Photo options in chat
+ Type as recipient...
Send
Recipient
Recipient sees revoked state
Recipient
I can see it now. Let me know if you want me to keep it private.
09:15
Reply
Forward
Download
Request access
viewing may already exist.
Revoked
No forwarding
Reviewable later
Download allowed
Access has been revoked for the recipient device.
Screenshots, saved copies, or prior viewing may already exist.
14:14
Photo options in chat
+ Type as recipient...
Send


Scenario 4 — Forwarding / secondary recipient
4
Secondary recipient
Forwarded-message recipient
Secondary recipient
Visible
Forwarding allowed
Reviewable later
Download allowed
Forwarded: Shared with revocable permissions.
Screenshots, saved copies, or prior viewing may already exist.
Restrictions from the original chat apply.
14:17
Photo options in chat
Forwarded photos from the recipient will appear here.
Secondary recipient
Forwarded-message recipient
Secondary recipient
ACCESS CHANGED
This photo is no longer visible.
The sender has revoked access in the current chat.
Screenshots, saved copies, or prior viewing may already exist.
Revoked
Forwarding allowed
Reviewable later
Download allowed
Access has been revoked for the recipient device.
Screenshots, saved copies, or prior viewing may already exist.
Restrictions from the original chat apply.
14:17
Photo options in chat
Forwarded photos from the recipient will appear here.

Scenario 5 — Co-subject / person depicted in the image

5



Appendix C

User Guide

C.1 Purpose

This appendix explains how to run the **Smart Photos** prototype on a local machine. Also a deployed version can be accessed at `https://smart-photos.vercel.app`. A concise, illustrated tour of every major flow is already given in Section 5.5.

C.2 Prerequisites

- **Node.js** version **20.x** (LTS recommended). The project uses TypeScript tooling aligned with Node 20; older major versions may fail to install or run dependencies.
- **npm** (bundled with Node.js). The instructions below use `npm`

C.3 Install and run

1. Open a terminal and change into the prototype folder:

   ```
   cd photo-revocation-prototype
   ```

2. Install dependencies:

   ```
   npm install
   ```

3. Start the development server:

   ```
   npm run dev
   ```

4. Open the default address printed in the terminal and make it full screen to have a better view.

C.4 Optional: production build

To run an optimised build (closer to a deployed Next.js app, useful for a stable demo):

```
npm run build
npm run start
```

C.5 Feature checklist (where to see each behaviour)

Feature	**What to do in the running prototype**
Pre-send permissions and screenshot warning	From the sender handset, open the photo picker, choose one or more images, then use *Send with revocable permissions*.
Post-send permission changes / revocation	On the sender handset, open the action menu on a sent photo message and use the post-send options (e.g. revoke access, forwarding, download, time limit). Expand the permission summary row under the bubble if needed.
Recipient view and revoked placeholder	Compare the recipient handset before and after sender-side revocation. Use recipient menu actions such as *Forward*, *Download*, or *Request access* when the design allows them.
View-once consumption	On the recipient handset, open a view-once photo via *Open*, then close the viewer and observe the consumed state on the bubble.
Forwarding and access request	Forward from a recipient handset when permitted. After revocation, use *Request access* and respond on the sender handset with *Accept* or *Reject*. Observe the forwarded-recipient panel for propagated state.
Co-subject flow	Use the co-subject handset panel. Read the notification card, open the photo menu, and use *Request revocation* where available. Complete *Accept*/*Reject* on the sender handset.

C.6 Optional: automated tests

```
npm run test:coverage
```

Further detail on testing is given in Section 5.6.

Appendix D

Ethical clearance

This appendix reproduces the ethics approval documentation issued for the user study reported in this dissertation.

Minimal Risk Registration Form

Section A: Confirm that your study requires KCL ethical clearance

Before completing a Minimal Risk Registration Form, you may find it useful to read through the accompanying guidance. Further advice for each question has been provided in information icons. Please ensure you have read these icons to confirm the eligibility of the project.

1 Is your study considered research as defined in the guidance icon information?
Please note studies deemed to be either a service evaluation or audit do not require ethical clearance.

- (•) Yes
- ( ) No

2 Does your study require external ethical review by either the Health Research Authority (which includes the NHS REC and Social Care REC) or the Ministry of Defence REC?
Please note - any research with NHS patients or NHS data will always require HRA review. Research with NHS staff-only may require HRA review. If you require HRA or NHS REC review, please select 'YES'. See guidance icon for further information on the HRA and MOD REC ethical review remit.

- ( ) Yes
- (•) No

3 Please indicate which of the following data collection methods your study involves:
Please note: You should only select the methods that you are certain will be employed for the project and you are able to outline in section 20(d).

- [x] Interviews
- [ ] Focus Groups/Workshop
- [ ] Questionnaires/Surveys/App based research tool
- [ ] Non-interventional classroom observations
- [ ] All other non-interventional observations
- [ ] Physical procedures (e.g taking body temperature, wearing a virtual reality headset, taking pulse)

Please note: Analysis of pre-existing data is not eligible for the Minimal Risk Registration Process. Before continuing, if your study involves the analysis of pre-existing data, please visit our 'Analysis of pre-existing data' page for advice on whether or not you will need to complete a 'Full Application Form' for ethical approval

Section B: Confirm that your study does not require High Risk review

IMPORTANT: Please note, the minimal risk self-registration process does not involve an ethics review conducted by the Research Ethics Office or the College Research Ethics Committee. As such it is the responsibility of the researcher and their supervisor (if applicable) to assess the potential risks involved to determine a projects eligibility. This should be done by carefully considering if each of the below questions may apply to the project.

Further information on of these potential risks can be found within the guidance icon of each question. If you select an answer that means the project is not eligible for minimal risk registration, you will be instructed complete a new full application form.

Minimal risk self-registration forms are subject to random audit. If a project is found to be ineligible during the audit process, ethical clearance for the project would not be valid and would be treated as research conducted without the correct level ethical clearance.

Before continuing, please confirm the following:

- [x] I understand the minimal risk self-registration process is not subject to review by the Research Ethics Office or the College Research Ethics Committee and I am responsible for assessing the potential risks in line with the guidance provided.

6 Does your study present any of the following risks to participants? (see information icon for guidance)

- [ ] a) Vulnerability: Does the study involve participants who are vulnerable, unable to give informed consent, or in a dependent position?
- [ ] b) Consent and deception: Will participants be asked to take part in the study without their informed consent or knowledge at the time, or will deception or an incomplete disclosure of any sort be involved?
- [ ] c) Participant disclosures: Could participants disclose details of their involvement in illegal activities or other activities that represent a threat to themselves or others or could lead to personal or professional reprisals (e.g. drug use, or professional misconduct)?
- [ ] d) Stress and anxiety: Could the study induce psychological stress or anxiety, or produce humiliation or cause harm or negative consequences beyond the risks encountered in a participant's usual, everyday life?
- [ ] e) Urgent mental health risks: Could participation in this research identify urgent mental health risks, including, but not limited to, suicidal ideation and/or self-harm intent?

You should only select the statement below if you have not selected any of the above risks. Your application will be invalid if you select the below statement in addition to any of the above.

- [x] I have answered no to all questions in the risk checklist above and I do not believe that my research is high risk

Section C: Confirm that your study does not require Low Risk review

7 Will any participants be under the age of 16?

- ( ) Yes
- (•) No

8 Will participation involve any subjects of the following nature?

Please note: The following points still apply if participation is anonymous. See information icon for further guidance.

- ☐ Subjects of a personal nature where it cannot be assumed that participants will feel at ease engaging with these topics without prior consideration
- ☐ Subjects that may not directly relate to lived experiences but do involve sensitive or controversial topics that you cannot assume a participant will be comfortable discussing
- ☐ Identification of low-level negative states. For example, the use of the Hospital Anxiety and Depression Scale (HADS)
- ☐ Details of personal medical history including past or current diagnoses or treatments
- ☑ None of the above

9 Do you have a current or prior relationship with any potential participants? (This includes professional and/ or personal relationships)

- ◉ Yes, I do have a current or prior relationships with potential participants.
- ○ No, I do not have any current or prior relationships with potential participants.

9a If you are in a position of influence or authority over potential participants, is there a possibility this could give rise to a perceived pressure to participate?

Please note: If you are directly involved in the teaching of participants this is considered a perceived pressure to participate.

- ○ Yes, there is potential for my existing relationship to give rise to a perceived pressure to participate.
- ◉ No, there is no potential for my existing relationship to give rise to a perceived pressure to participate.

10 Gatekeeper Permission: Will you require an individual or organisation to grant you permission to approach/ access your intended participants? This includes gatekeepers contacting participants on your behalf.

Please note: if you will be attempting to access any closed online groups which require admin permission (such as private Facebook groups), this is considered to be use of a gatekeeper.

- ○ Yes, I will be using a gatekeeper to access potential participants
- ◉ No, I will not be using a gatekeeper to access potential participants

11 Can you confirm that the participants will not be subjected to any undue incentives to participate? (Receipt of a project report or a reasonable reimbursement such as travel/lunch costs or a voucher would not be considered undue incentives.)

- ○ Yes, participants will be subject to undue incentives
- ◉ No, participants will not be subjected to any undue incentives

12 You have determined that your project is eligible for minimal risk registration. Before completing your minimal risk registration form you must confirm that you have read the Minimal Risk Guiding Principles at the link below and that you will adhere to these principles in the conduct of your research.

Minimal Ethical Risk Guiding Principles

- ☑ I confirm that I have read the Minimal Risk Guiding Principles and will adhere to the principles within.

Section D: General Information

13 Applicant Details

Title	First Name	Surname
Mr	Mert	Ayranci

Department: Department of Informatics

Email: mert.ayranci@kcl.ac.uk

14 Faculty/Institute/School

Please refer to the information icon if you are unsure of your Faculty/Institute/School.

Natural and Mathematical Sciences

15 Applicant Status

Undergraduate

15 Supervisor Details

a)

Title	First Name	Surname
Dr	Kovila	Coopamootoo

Department: Department of Informatics

Email: kovila.coopamootoo@kcl.ac.uk

16 Are there any other investigators/collaborators involved in the study?

- ○ Yes
- ◉ No

Section F: Study Overview

18 Project Title

A working title for your project that accurately reflects its aims

Designing For Consent Revocation in online messaging platforms (WhatsApp Photo Sharing)

19 Anticipated start date for the collection of data: 20/03/2026

20 Study Overview: Briefly describe your research project, as you would to a potential participant, highlighting the aims of your research, who your participants are (type of people rather than individual names), how you will recruit them and what will be asked of them (try to use no more than 50 words for each section).

a) What are the aims of your study?

You should explain what the principal research question is and the specific objectives of the study

The principal research question of this study is: How can WhatsApp be designed to support clear, usable, and trustworthy mechanisms for revoking or modifying consent after users share photos in private conversations? The study focuses on the problem that, once a photo is sent, users often lose meaningful control over it, even though consent and comfort can change over time. The specific objectives are to examine the limitations of WhatsApp's current photo-sharing design, investigate users' expectations and concerns about post-sharing control, define requirements for effective consent-revocation features, design and prototype possible interface solutions, and then evaluate whether those proposed features are understandable, usable, trustworthy, and fair.

b) Who are your potential research participants?

This should outline any specific criteria participants must meet in order to be eligible take part (e.g. age, ethnicity, gender, members of a specific group etc.)

Potential participants for this study are adults aged 18–24 who are active users of mobile messaging platforms, particularly apps used for sharing photos in private conversations, such as WhatsApp. Participants should have regular experience of sending or receiving photos through these platforms so that they can meaningfully comment on the prototype and the scenarios shown during the interview.

c) How will participants be identified and approached?

This must address each data collection method selected in Q3. For example, if approaching potential participants in person you should explain the circumstances in which this will take place or if approaching potential participants by email you should explain how you will obtain email addresses.

Potential participants will be identified through purposive sampling. The study will recruit adults aged 18–24 who are active users of mobile messaging platforms and have experience sharing photos in private chats. Participants will be approached through the researcher's personal and university networks, for example by sending a short invitation message by email or direct message, or by sharing a recruitment message in relevant student or social groups.

d) What will participation involve?

Briefly explain how each data collection method, indicated in Q3, will be used to collect data. This should include what participants will be asked to do at each stage of the project.

Please note: Failure to address each method indicated in Q3 will result in your form being invalidated.

Participants will take part in a one-to-one semi-structured online interview. Before the interview starts, the researcher will explain the study verbally, answer any questions, and obtain verbal informed consent. With permission, the interview will be audio-recorded. Participants will first answer a few warm-up questions about messaging and photo-sharing practices, and will then be shown screenshots from a prototype. They will be asked to discuss what they think the screens mean, how they interpret the features, and how they feel about them. Follow-up questions will explore clarity, fairness, trust, and perceived control. At the end, participants will be asked a few final reflection questions. They may skip any question or stop participation at any time.

e) Provide example questions participants may be asked:

Provide an example of the types of questions participants may be asked for each data collection method selected in Q3. These examples should demonstrate the nature of the topics that will be discussed but does not need to be an exhaustive list.

Participants will be asked questions about their current use of messaging apps and photo-sharing practices, as well as their reactions to a series of prototype screenshots. Example questions include: “Can you tell me how you usually share photos on apps like WhatsApp?”, “Have you ever wanted more control over a photo after sending it?”, “What do you think is happening on this screen?”, “What do you think these settings mean?”, “How would you feel if access to a shared photo changed after it was sent?”, “Does this feel fair to both sender and recipient?”, and “Would a feature like this make you feel more in control or trust the platform more?” These examples are illustrative and not exhaustive.

21 Confirm which of the following consent processes will be used:

- ☐ Written Consent: Written consent will be recorded in either paper or electronic form in advance of participation. A written description of the research will also be provided to all potential participants.
- ☑ Verbal Consent: I am able to demonstrate that written information and consent is not practical, or not appropriate, so I confirm that I will follow College guidance on providing information and gaining consent from participants verbally.
- ☐ Anonymous submission of survey/questionnaire/app based research tool data: A written description of the research will be provided to all potential participants and it will be made clear that the submission of a completed survey/questionnaire/app data implies consent.
- ☐ Non-invasive observations that do not involve any interaction with participants and no identifying information will be recorded.

21a Explain why written consent is not practical or appropriate.

Written consent is not considered practical for this study because participation takes place remotely through online interviews, and the project involves minimal risk. Instead, the study will be explained verbally at the start of each interview, participants will have the opportunity to ask questions, and verbal informed consent will be obtained before any data is collected

◉ Provisions for verbal consent: I confirm that I will follow KCL guidelines on gaining verbal consent and wherever practically possible I will provide all potential participants with appropriate written study information. This should include researcher contact details for purposes of questions, complaints or withdrawal requests.

Please note, where possible when obtaining verbal consent participants should be provided with a written information sheet. You should also ensure you have created a record of the consent process.

KCL guidelines on gaining verbal consent and template information sheets

22 Does the project involve the collection and/or use of personally identifiable information (as outlined in UK GDPR)?

Identifiable information is data that can be used to identify an individual, either directly (such as full name, address, Twitter handle, etc) or indirectly through the combination of several pieces of data. The most common examples are names, contact details, audio/ video recordings, usernames etc. However, data that has the potential to indirectly identify a participant should also be treated as identifiable.

Please see the guidance icon for more examples of when data should be considered identifiable or contact the Research Governance Office: rgo@kcl.ac.uk

Please indicate which of the following applies:

- ○ Yes, the project involves the collection and/or use of identifiable information
- ◉ No, all data I will collect or access for this research will be completely anonymous. It will not be possible for me to identify an individual(s) at any point in the project.

PLEASE NOTE:

- **If you are using email addresses to contact participants you should select 'Yes' under question 22 above.**
- **If you are asking participants to sign consent forms with their name and/or signature you should select 'Yes' under question 22 above**

◉ I understand that it is the responsibility of the researcher to ensure that all research data is appropriately handled and stored during and after the study in compliance with College guidelines:

King's research data management guidelines and King's Data Retention Schedule guidance

Section G: Declaration and Signatures

23 Researcher/Applicant Declaration:
IMPORTANT NOTE FOR STUDENTS: Please ensure that you have signed the form in this section before requesting your Supervisor's signature in section 24 below.

By signing this form I confirm the following:

- The information supplied above is to the best of my knowledge accurate.
- I understand the minimal risk self-registration form is not subject to review by CREC or the REO, and it is my responsibility to ensure the project meets the minimal risk eligibility criteria based on the guidance provided.
- I have read the Minimal Ethical Risk Guiding Principles and clearly understand my obligations and the rights of participants, particularly as regards obtaining informed consent.
- The participant selection and recruitment procedures, including the recruitment documents to be provided and the manner of obtaining informed consent, are appropriate and the ethical issues arising from the project have been considered (and agreed with my Supervisor if applicable).
- I understand that I must not commence research with human participants until I have received confirmation of minimal risk registration.

Signed: This form was signed by Mert Ayranci (mert.ayranci@kcl.ac.uk) on 20/03/2026 16:26

24 Supervisor Declaration:

Once you have electronically signed the form in section 23, you should then request your Supervisor's signature using the below 'Request' button. Once your Supervisor has also signed the form the form will be automatically submitted for registration.

Please note: Following submission for registration you must wait until you have received a letter from REMAS confirming that your project has been registered before commencing data collection.

Request Supervisor's Signature:	**Signed:** This form was signed by Kovila Coopamootoo (kovila.coopamootoo@kcl.ac.uk) on 23/03/2026 12:18

Appendix E

Source Code

Authorship statement

I verify that I am the sole author of the programs contained in this folder, except where explicitly stated to the contrary.

Signed: Mert Ayranci **Date:** April 20, 2026

E.1 Submission contents

The complete source code is submitted alongside this report as a separate ZIP file. This appendix is intended as a navigation aid for that archive rather than a verbatim listing. It mirrors the structure of the prototype repository and gives a one-line description of every file or unit.

The prototype is a Next.js (App Router) front-end written in TypeScript, styled with Tailwind CSS, and tested with Vitest and React Testing Library. Build and run instructions are provided in Appendix C; the architectural narrative is given in Chapter 5.4.

E.2 Repository layout

At the top level the archive is organised as follows:

Path	Purpose
`src/app/`	Next.js App Router entry point, root layout, and global stylesheet.
`src/components/photo-revocation-demo/`	React components implementing every interface region of the prototype.
`src/lib/photo-revocation-demo/`	Domain types, helper functions, constants, and initial in-memory state.
`src/lib/`	Cross-cutting permission logic shared between components and tests.
`tests/`	Vitest suite organised into `unit/`, `components/`, `app/`, and shared `helpers/`.
Root configuration	TypeScript, Next.js, Vitest, ESLint, Tailwind PostCSS, and package manifest files.

E.3 File index

The following table lists every file in the submitted archive, grouped by directory, with a one-line description of its responsibility.

Application entry (`src/app/`)

File	Description
`page.tsx`	Thin route component that mounts the `PhotoRevocationDemo` orchestrator on the root URL.
`layout.tsx`	Root layout, metadata, and font configuration applied to every page.
`globals.css`	Tailwind CSS entry point and global style resets.

Domain logic (`src/lib/`)

File	Description
`photoModel.ts`	Pure permission model: `computeEffectiveStatus`, `forwardingAllowed`, `formatExpiryCountdown`, and time-limit helpers.
`photo-revocation-demo/types.ts`	TypeScript types for messages, permissions, participants, and modal state (single source of truth for the data model).
`photo-revocation-demo/constants.ts`	Static configuration: scroll-wheel sizing and the preset photo album.
`photo-revocation-demo/initialState.ts`	Default draft permissions, default time limits, initial conversation, and send-history seeds.
`photo-revocation-demo/helpers.ts`	Derivation helpers used by the UI: `getEffectivePermissions`, `getSourcePolicyPermissions`, view-once consumption checks, and download utilities.

React components (`src/components/photo-revocation-demo/`)

File	Description
`PhotoRevocationDemo.tsx`	Central orchestrator: owns all shared state, handlers, and the four-handset layout.
`PhoneFrame.tsx`	Renders the device chrome for each stakeholder view and provides the per-handset overlay context.
`chatPanelPortalContext.tsx`	React context exposing portal and scroll references so overlays render inside the correct handset.
`MessageList.tsx`	Iterates a message array and dispatches each entry to a text or photo bubble.
`TextMessageBubble.tsx`	Renders text, reply previews, and accept/reject controls for access and revocation requests.
`PhotoMessageBubble.tsx`	Renders photo messages, the collapsible permission summary, role-aware action menu, and revocation submenu.
`PhotoPreview.tsx`	Renders a single photo asset with placeholder, view-once, and revoked variants.
`PhotoStackGrid.tsx`	Multi-photo collage layout for messages with several attached assets.
`PhotoStackLightbox.tsx`	Full-screen gallery viewer that portals into the active handset frame.
`ComposerView.tsx`	Pre-send composer with caption input, permission chips, and the screenshot warning.
`PickerSheet.tsx`	Photo picker sheet with multi-select and the dual *Send*/*Send with revocable permissions* actions.
`ScrollWheelColumn.tsx`	Reusable wheel control used by the time-limit picker.
`TimeLimitModal.tsx`	Post-send modal for adjusting a photo's expiry.
`RequestAccessPanel.tsx`	Inline footer panel for recipients to compose an access request.
`RequestRevocationPanel.tsx`	Inline footer panel for co-subjects to compose a revocation request.
`ViewOnceInChatModal.tsx`	Full-screen viewer that handles the view-once consumption lifecycle.
`ViewOnceSingleTick.tsx`	Small SVG indicator marking a view-once photo as consumed.
`PrototypeStateSummary.tsx`	Side panel exposing the live canonical permission record for evaluator inspection.

Tests (`tests/`)

File names below are listed relative to the `tests/` root and are grouped by sub-directory.

File	Description
`unit/photoModel.test.ts`	Unit tests for the pure permission model and time-limit helpers.
`unit/helpers.test.ts`	Unit tests for derivation helpers, including effective-status fallbacks.
`components/ComposerView.test.tsx`	Component tests for the caption input, permission chips, and the warning banner.
`components/PickerSheet.test.tsx`	Component tests for multi-select and the two send actions.
`components/PhotoMessageBubble.test.tsx`	Branch tests for active, revoked, view-once, and download-disabled variants.
`components/PhotoStackGrid.test.tsx`	Layout tests for multi-photo collages.
`components/PhotoStackLightbox.test.tsx`	Tests for the portalled full-screen gallery.
`components/RequestPanels.test.tsx`	Tests for both inline request panels (access and revocation).
`components/ScrollWheelColumn.test.tsx`	Tests for the duration picker primitive.
`components/TimeLimitModal.test.tsx`	Tests for applying and cancelling post-send expiry changes.
`components/TextMessageBubble.test.tsx`	Tests for replies and request response controls.
`components/ViewOnceInChatModal.test.tsx`	Tests for view-once open, close, and consumption state.
`components/ViewOnceSingleTick.test.tsx`	Snapshot test for the consumed indicator.
`components/PrototypeStateSummary.test.tsx`	Tests that the state panel reflects the canonical record.
`components/PhotoRevocationDemo.test.tsx`	Integration-style journeys covering revocation, forwarding, and request flows across handsets.
`app/page.test.tsx`	Smoke test that the route mounts the orchestrator without error.
`helpers/photoDemoUi.tsx`	Shared Testing Library locators and assertions reused across integration tests.

Documentation (`docs/`)

File	Description
`README.md` (root)	Quick-start instructions, architectural overview, and test commands.

Configuration (root)

File	Description
`package.json`	Project manifest, dependency declarations, and `npm` scripts (`dev`, `build`, `start`, `test`, `test:coverage`, `lint`).
`package-lock.json`	Resolved dependency lockfile for reproducible installs.
`tsconfig.json`	TypeScript compiler configuration including the `@/*` path alias.
`next.config.ts`	Next.js framework configuration.
`next-env.d.ts`	Auto-generated Next.js TypeScript ambient declarations.
`vitest.config.ts`	Vitest configuration: jsdom environment, alias resolution, and coverage settings.
`vitest.setup.ts`	Test setup including `@testing-library/jest-dom` matchers.

E.4 Notes

- All work on the prototype stopped once the archive was uploaded to KEATS. Several copies of the submitted version are retained locally and one of these copies will be used in the project examination.
- The deployed demonstration at `https://smart-photos.vercel.app` is built from the same submitted source and is provided only as a convenience; the canonical artefact for marking is the submitted ZIP archive.
- No third-party code is included in the archive beyond declared `npm` dependencies, which are installed at build time from `package.json` and `package-lock.json`.